ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

# MATHEMATICS

## PAPER 1 · PRACTICE B · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central Grade 8 Mathematics paper 1 · practice b.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct answers.

## QUESTION 1

**1** 1.1 C ✓ 1.2 A ✓ 1.3 D ✓ 1.4 B ✓ 1.5 D ✓ 1.6 B ✓ 1.7 C ✓ 1.8 A ✓ 1.9 B ✓ 1.10 D ✓ [10]

## QUESTION 2

**2.1** $2 + 3 + 5 = 10$ parts; R2 400 ÷ 10 = R240 per part (3)
Ayanda R480 ✓ Bongi R720 ✓ Chris R1 200 ✓

**2.2** (a) $360 \div 100 \times 7{,}5$ ✓ $= 27\ \ell$ ✓ (b) $27 \times$ R23,40 = R631,80 ✓ (3)

**2.3** (a) $15\% \times$ R600 = R90 ✓ → R690 ✓ (4)
(b) $20\% \times$ R690 = R138 ✓ → R690 − R138 = R552 ✓

**[10]**

## QUESTION 3

**3.1** (a) $36 \div (-4) + (-27)$ ✓ $= -9 - 27$ ✓ $= -36$ ✓ (5)
(b) $12 - (-5)$ ✓ $= 17$ ✓

**3.2** (a) $a^8 \div a^4$ ✓ $= a^4$ ✓ (b) $9x^4y^2$ ✓✓ (c) $8x^3$ ✓ (5)

**[10]**

## QUESTION 4

**4.1** (a) $^{13}/_4 - {}^{11}/_6$ ✓ $= {}^{39}/_{12} - {}^{22}/_{12} = {}^{17}/_{12} = 1\,{}^5/_{12}$ ✓✓ (7)
(b) $^8/_3 \div {}^4/_3 = {}^8/_3 \times {}^3/_4$ ✓ $= 2$ ✓
(c) $^5/_6 + {}^3/_6$ ✓ $= {}^8/_6 = 1\,{}^1/_3$ ✓

**4.2** 3,142 ✓ (1)

**[8]**

## QUESTION 5

**5.1.1** $T_n = 2n + 1$ ✓✓ (2)

**5.1.2** $2(50) + 1 = 101$ ✓ (1)

**5.1.3** $2n + 1 = 81 \rightarrow 2n = 80$ ✓ → Figure 40 ✓ (2)

**5.2** (a) $8x - 12 - 3x + 15$ ✓✓ $= 5x + 3$ ✓ (8)
(b) $x^3 - 2x^2$ ✓ $+ 3x$ ✓
(c) $3x^2y$ ✓ $- 1$ ✓
(d) $4x^4$ ✓

**5.3** (a) $2x = 14$ ✓ $\rightarrow x = 7$ ✓ (6)
(b) $x - 1 = 12$ ✓ $\rightarrow x = 13$ ✓
(c) $x^2 = 25$ ✓ $\rightarrow x = 5$ or $x = -5$ ✓

**[19]**

## QUESTION 6

**6.1** $3x + 10° = 2x + 30°$ (vertically opposite angles) ✓ $\rightarrow x = 20°$ ✓ (4)
$\angle AOC = 3(20°) + 10° = 70°$ ✓ $\angle COB = 180° - 70° = 110°$ (adjacent angles on a straight line) ✓

**6.2** (a) $\angle ACB = 180° - 115° = 65°$ (adjacent angles on a straight line) ✓ (4)
(b) $\angle B = \angle ACB = 65°$ (base angles of an isosceles triangle, AB = AC) ✓✓
(c) $\angle A = 180° - 65° - 65° = 50°$ (sum of the angles of a triangle) ✓

**6.3** $x + 2x = 180°$ (co-interior angles, AB ∥ CD) ✓ $\rightarrow 3x = 180°$ ✓ $\rightarrow x = 60°$ ✓ (3)

**6.4** $^{EF}/_{BC} = {}^{DE}/_{AB} \rightarrow EF = 8 \times {}^{9}/_{6}$ ✓ $= 12$ cm ✓ (2)

**6.5** Square ✓ (1)

**[14]**

## QUESTION 7

**7.1** $d^2 = 80^2 + 60^2 = 10\,000$ ✓ $\rightarrow d = 100$ cm ✓ (2)

**7.2** $x^2 = 13^2 - 5^2 = 144$ ✓ $\rightarrow x = 12$ cm ✓ (2)

**7.3** (a) $A = 3{,}14 \times 10^2$ ✓ $= 314$ cm$^2$ ✓ (4)
(b) ½ × 2 × 3,14 × 10 + 20 ✓ = 31,4 + 20 = 51,4 cm ✓

**7.4** (a) $V = 60 \times 30 \times 40$ ✓ $= 72\,000$ cm$^3$ ✓ $= 72\ \ell$ ✓ (6)
(b) Base $60 \times 30 = 1\,800$; front and back $2(60 \times 40) = 4\,800$; sides $2(30 \times 40) = 2\,400$ ✓✓
Total $= 9\,000$ cm$^2$ ✓

**[14]**

## QUESTION 8

**8.1.1** A'(−1 ; −1) ✓ B'(−4 ; −1) ✓ C'(−1 ; −3) ✓ (3)

**8.1.2** (−1 ; 3) ✓ (1)

**8.1.3** $(12\ ;\ 3)$ ✓ (1)

**[5]**

## QUESTION 9

**9.1.1** 2 books ✓ (1)

**9.1.2** $0(4) + 1(7) + 2(9) + 3(6) + 4(4)$ ✓ $= 0 + 7 + 18 + 18 + 16 = 59$ books ✓ (2)

**9.1.3** $59 \div 30 = 1{,}97$ books ✓ (1)

**9.1.4** The median lies between the 15th and 16th values ✓; the first 11 learners read 0 or 1 book and learners 12 to 20 read 2 books, so the median is 2 ✓ (2)

**9.1.5** $4 - 0 = 4$ ✓ (1)

**9.1.6** ${}^{6+4}/_{30} = {}^{10}/_{30}$ ✓ $= {}^{1}/_{3}$ ✓ (2)

**9.1.7** A bar graph ✓ (1)

**[10]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**